



High Performer Attrition Analysis

Kansora Medical

Case study scenario

This project uses the fictional IBM HR Analytics Employee Attrition dataset. Kansora Medical is a fictional Osaka-based medical-device manufacturer with approximately 5,000 employees.

In this scenario, Kansora Medical's People Analytics team has been asked to investigate attrition among high performers and identify priorities for improving retention.

All company details and business assumptions are fictional and used solely for analytical illustration.

Background and Objectives

Business objective

Reduce unwanted attrition among high-performing Kansora Medical employees by identifying the factors most strongly associated with employees leaving and prioritising areas for management action.

Initial questions

- Where is attrition concentrated across the workforce?
- Which factors are most strongly associated with higher or lower attrition odds after controlling for other workforce characteristics?
- Which of these factors are sufficiently strong and actionable to inform retention priorities?
- Can attrition be predicted at an individual level using the available workforce data, and would such predictions be appropriate to use in practice?

Analysis Scope

- 1,470 high-performing employees (rated Excellent or Outstanding)
- These high performers represent 30% of the total employee population
- Attrition measured over the previous 12 months

Prior Research on Retention

Prior research suggests that employee attrition is influenced by a combination of work design, employee attitudes, management experience and development opportunities, not compensation alone.

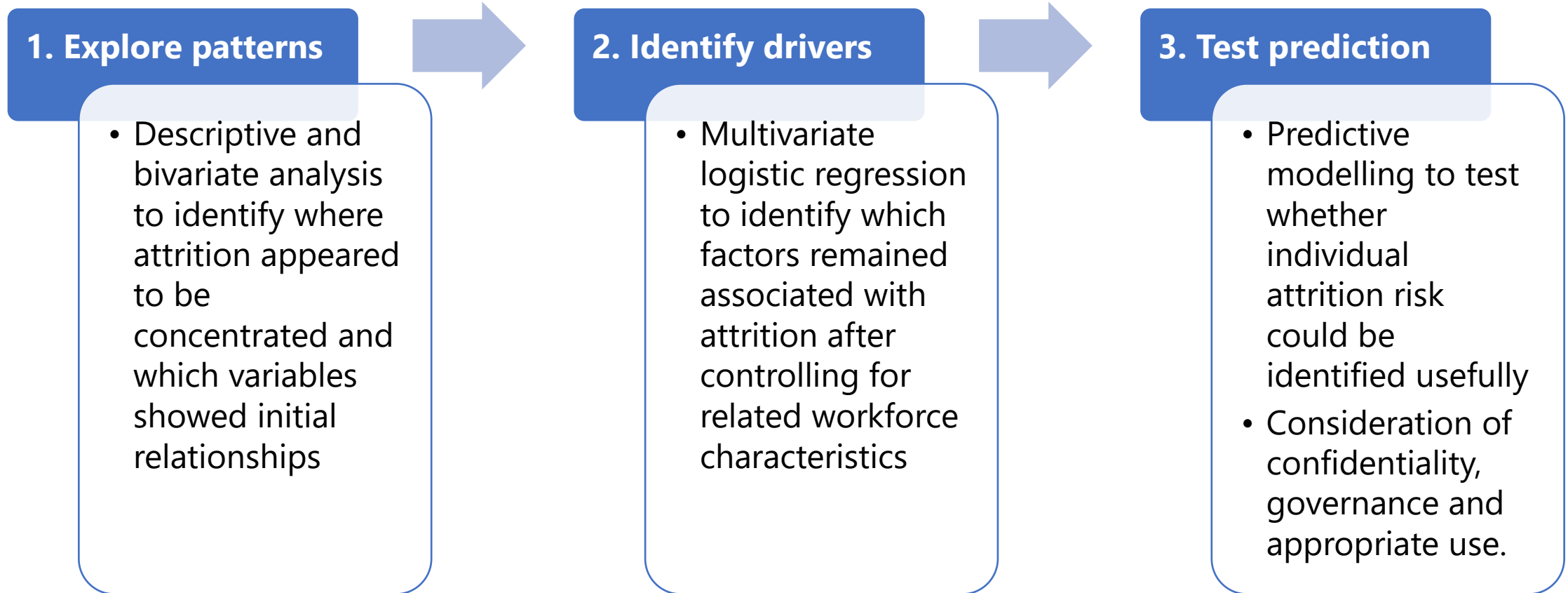
Key Research Findings

- **Attrition is multi-causal:** Manager behaviour, job characteristics, team climate, rewards, stress and employee attitudes can all contribute, so diagnosis should come before intervention.
- **Job satisfaction and organisational commitment matter:** Employees who are more satisfied and committed are generally less likely to leave.
- **The nature of the job matters:** Autonomy, feedback, meaningful work and other job characteristics are associated with lower attrition.
- **Pay is not the whole story:** Compensation has only a small to moderate relationship with attrition, while development opportunities, training and other rewards may matter more.



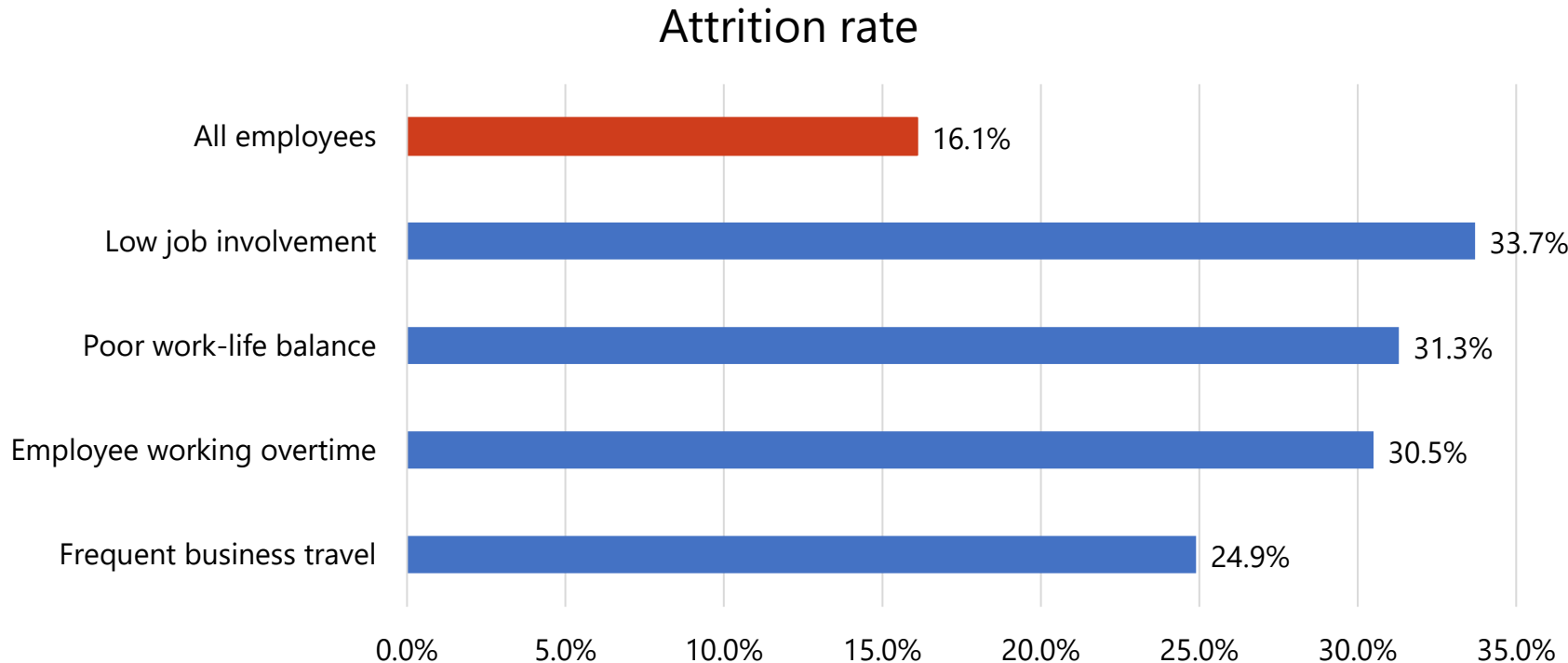
Source: [Science for Work evidence summaries on employee turnover](#)

Analytical Approach



1. Explore Patterns

Initial analysis showed that attrition was substantially above the 16% workforce average among employees reporting low involvement or poor work-life balance, and among those working overtime or travelling frequently.



Note: These are unadjusted bivariate comparisons. The subsequent multivariate analysis tests which relationships remain after controlling for other workforce characteristics.

2. Identify Drivers

Odds ratios (OR) show the adjusted association with attrition: above 1 = higher odds, below 1 = lower odds. The strongest patterns are around work demands and employee experience.

Theme	Lower attrition odds	No clear independent effect	Higher attrition odds
Work demands & balance	Better work-life balance (OR 0.24-0.37 vs Bad)	—	Overtime: Yes (OR 8.60) - Frequent business travel (OR 8.30)
Employee experience	Very high job satisfaction (OR 0.49)	—	- Low job involvement (OR 5.02) - Low environment satisfaction (OR 3.31) - Low relationship satisfaction (OR 2.67) - Low job satisfaction (OR 1.97)
Career & development	- Longer time in current role (OR 0.86 per year) - Longer time with current manager (OR 0.87 per year) - More training (OR 0.84 per event)	Performance rating: No significant difference between Excellent and Outstanding - Most job-role effects weakened after adjustment	- More companies worked (OR 1.23 per company) - More years since last promotion (OR 1.19 per year)
Compensation & rewards	—	Neither percent salary hike nor stock option level had a significant independent association	—

Note: Continuous predictors are shown per stated unit. These are adjusted associations, not causal effects.

3. Test Prediction

The models identified some attrition signal, but did not beat the 84% “all stay” accuracy baseline. Individual-level risk scores also raise confidentiality and governance concerns, so leader use is not recommended.

Model	Overall accuracy	Leave recall	Leave precision
Logistic regression	83.7%	39%	49%
Decision tree	69.8%	48%	26%

Metric definitions:

- Leave recall = % of actual leavers correctly identified
- Leave precision = % of employees predicted to leave who actually left.

Recommendations

Business leaders should work with their local HRBP to turn these findings into practical, team-specific actions to strengthen high-performer retention.

Theme	Action
Work demands & balance	• Prioritise persistent overtime as a retention risk • Review and reduce unnecessary or excessive business travel • Address workload and work-life balance in high-risk areas
Employee experience	• Focus action planning on job involvement and satisfaction • Use survey results to identify local drivers and agree targeted actions • Strengthen follow-through and accountability for engagement plans
Career & development	• Support employees through role and manager transitions • Maintain strong training and development opportunities
Compensation & rewards	• Avoid over-relying on salary increases or stock options as a retention lever

Next Steps

Using the high-performer attrition findings, HR and business leaders should translate the recommendations into relevant local actions and review progress over time.

Timing	Action	Ownership
Week 0 - Today	Review findings, agree priorities and confirm rollout approach	People Analytics + HR leadership
Week 1	Brief HRBPs on findings, interpretation and their role in supporting business action	People Analytics
Weeks 2-3	Share key findings with business leaders	People Analytics + HRBPs
Weeks 3-6	Identify local priorities and agree targeted actions	Business leaders + HRBPs
Quarter +1	Review progress on agreed actions and relevant indicators	HRBPs + business leaders
Next survey / attrition cycle	Reassess whether indicators and attrition patterns have changed	People Analytics

Conclusion

Summary

- Attrition is most strongly associated with **work demands and employee experience**, particularly overtime, frequent travel, work-life balance, involvement and satisfaction.
- **Compensation measures** such as salary increases and stock options showed little independent effect after controlling for other factors.
- Management should focus on **targeted local action**, with business leaders and HRBPs using the findings to address the most relevant retention risks in their teams.
- Predictive modelling showed some signal, but **individual-level use is not recommended** because performance was limited and governance concerns remain.

Limitations

- Findings show **association, not causation**
- Some survey-related measures lack detailed definitions
- Predictive performance for actual leavers was limited

